

GYROSCOPIC CUP HOLDER

Design Development • CAD Modeling • Prototyping & Evaluation



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1. Project Overview

The Gyroscopic Cup Holder is a self-leveling beverage mount designed to keep a drink upright and spill-free while its supporting platform is in motion. The device is aimed primarily at cyclists and wheelchair users, though its design generalizes to any application where a cup must remain stable on a moving surface — from car interiors to strollers and boats.

Stability is achieved through a two-ring gimbal assembly: an inner ring cradles the cup while an outer ring allows the inner ring to pivot freely about a second axis. As the mounting platform tilts, pitches, or rolls, gravity keeps the cup itself oriented upright within the concentric rings, preventing sloshing and spills without any electronics or active control.

Market Reference

Before finalizing a concept, three comparable products already on the market were reviewed to benchmark form factor, mounting style, and mechanism type.



Gimbal-style ring holder (reference)



Clamp-mounted rotating holder (reference)



Compact stroller-mount holder (reference)

2. Design Requirements

The following requirements were developed to guide the concept and shaped every downstream decision in geometry, material, and cost.

Use & Durability

- Reduce spills and provide stabilization during normal, everyday motion — in a car, on a bike, or on a wheelchair.
- Intended for repeated, long-term use rather than a single occasion; the holder should become a daily-carry item.
- Built to survive thousands of use cycles: easy to clean, impact-tolerant, and stable under load.

Portability & Mounting

- Transportable and quickly adjustable — removable from a bike when not needed, or semi-permanently fixed to a car via small clamps or screws.

Size, Weight & Finish

- Sized for standard cups (under 80 mm in diameter) without adding excessive bulk.
- Total weight kept under 1 kg for comfortable use.
- Low-profile, purpose-driven finish in neutral or matte-black tones to complement car interiors; an optional rubberized contact surface improves grip.

Cost & Market Positioning

- Student prototype cost constrained by NJIT Makerspace 3D-printing access, targeted under \$50.
- A production-ready version is targeted at \$25–\$30 retail, based on a competitive scan of similar products on Amazon and Etsy ranging from \$10 to \$50.

Maintenance

- Low-maintenance, easy-to-clean materials; simplicity was prioritized over ornamentation. A premium, rougher-terrain variant (e.g., for boats or golf carts) is a possible future tier.

3. Design Criteria & Constraints

- Total weight must not exceed 1 kg for comfortable, everyday handling.
- The device is passive by default; if a powered variant is pursued, it is limited to a small low-voltage system (5 V, small battery).
- Prototype cost is targeted under \$100 for students, and under \$50 for consumers in a finished product.
- Must accommodate cups of varying diameters, ideally through an adjustable holder.
- Mounting must be simple and comfortable, using clamps or screws as needed.

4. Design Goals

- Fit standard cup sizes.
- Mount and remove easily.
- Prevent spillage and sloshing during motion.
- Use materials that are easy to clean and maintain.
- Meet the target cost requirements.
- Operate safely and quietly, without drawing attention.

- Remain simple to manufacture.
- Minimize user interaction — lift the cup out and set it back in.

5. Functional Requirements

The table below details the functional requirements the design must satisfy to operate effectively.

#	Requirement
1	Hold cups of standard size diameters, preferably 64–100 mm.
2	Maintain the cup opening within a couple of degrees relative to the platform's tilt.
3	Allow full continuous rotation (if necessary) and several tens of degrees of pitch and roll.
4	Provide a non-slip, shock-absorbent interface.
5	Mount to common surfaces — bike handlebars, a car dashboard, or rail clamps — via studs or straps.
6	Resist liquid exposure and remain easy to wash and clean.
7	Operate passively, without a battery; if powered, limited to 5–12 V and under 2 A.
8	Tolerate a wide temperature range, roughly -10°C to 50°C , from iced drinks to hot coffee.

6. Design Constraints

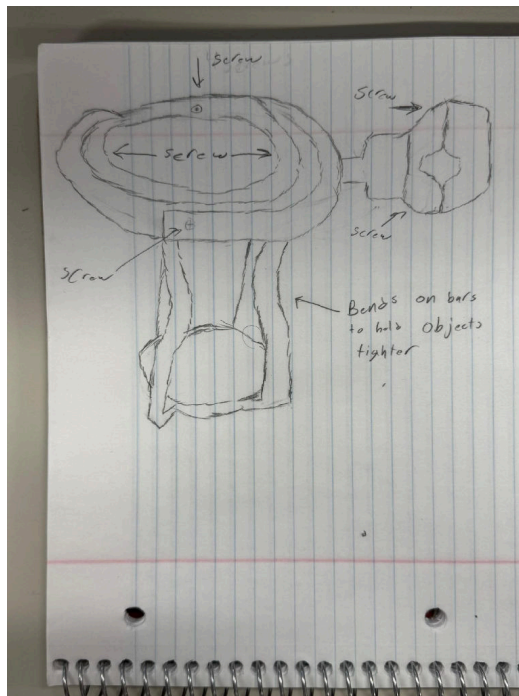
#	Constraint
1	Weight & Size — must stay compact and light for standard mounts; excess size or weight reduces stability.
2	Cost — higher-quality materials and components raise the price, which can limit affordability.

7. Conceptual Design

Two related concepts were sketched and evaluated before committing to a CAD model.

Concept 1

A gimbal formed from two concentric rings, joined by screws, with the cup cradle supported by bent bars for a tighter hold. The mount arm attaches to the outer ring via a separate clamp module.



Original concept sketch — gimbal ring assembly with cup cradle and mounting clamp.

Concept 2

A refinement of Concept 1 in which the entire cup-cradle base is molded directly to the support bars, simplifying assembly by reducing the fastener count.

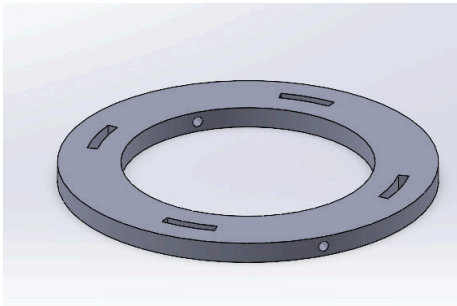


Visual reference used to guide the ring-and-cradle geometry.

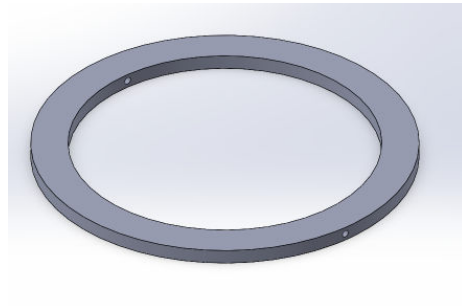
8. CAD Modeling

The concepts were developed into full 3D models in SolidWorks. Work was divided by component, with each teammate modeling and refining specific parts before final assembly.

Individual Components



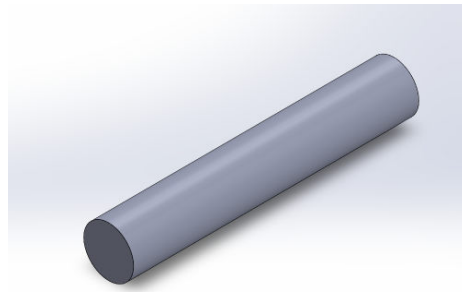
Inner ring — modeled by Ramirez



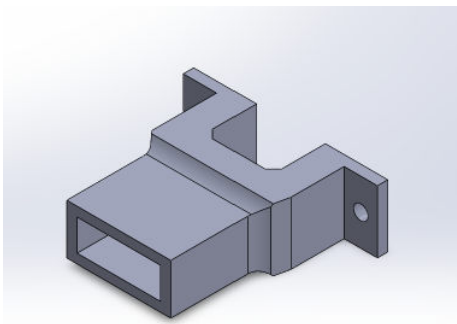
Outer ring — modeled by Ramirez



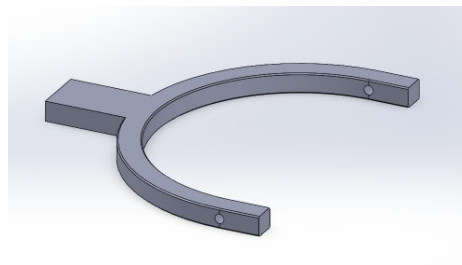
Support ribs — modeled by Mustafa



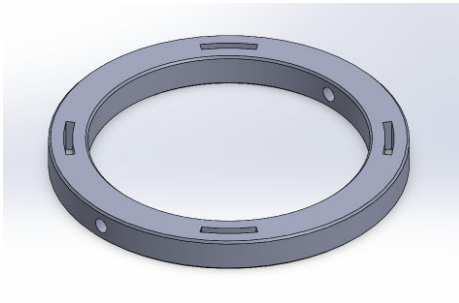
Cup-cradle base with ribs



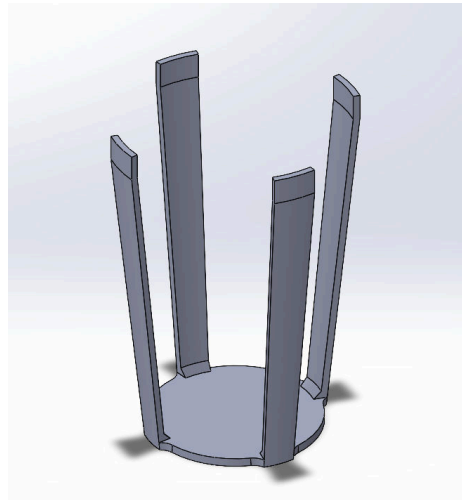
Handlebar clamp / mount block



Adjustable C-clamp collar



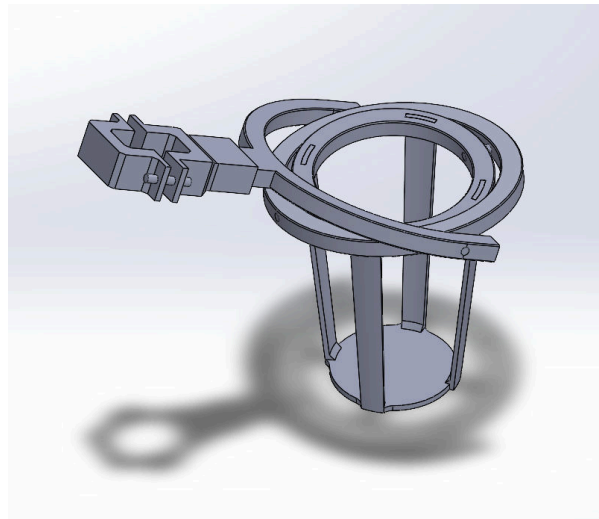
Ring detail — fastener bosses and slots



Connecting pin / rod

Assembly

The individual components were combined into a single assembly to verify fit, ring clearance, and range of motion before the parts were sent to print.



Full assembly — gimbal ring pair, support ribs, and mounting clamp (rough draft).

Exploded View

An exploded view was generated to document part relationships and fastening order ahead of manufacturing.



Exploded assembly showing the clamp, ring pair, connecting pins, and rib base.

9. Manufacturing & Testing

3D Printing

All parts were printed in PLA across multiple printers. Given the compact geometry, most components printed in a couple of hours each; the longest single piece — the support ribs — took just over an hour. Limited printer availability meant parts were produced in batches over several sessions. Total manufacturing cost for the prototype came to approximately \$10.



Assembled 3D-printed prototype.



Prototype holding a standard water bottle.

Prototype Evaluation

Print Quality

All features printed correctly. The primary issue was the absence of any dimensional tolerance in the original models: mating parts fit very tightly, and several — particularly the support ribs — required light sanding to assemble. Leftover PLA residue in the ring pin-holes also had to be cleared, in some cases using a screwdriver to push the material free.

Durability

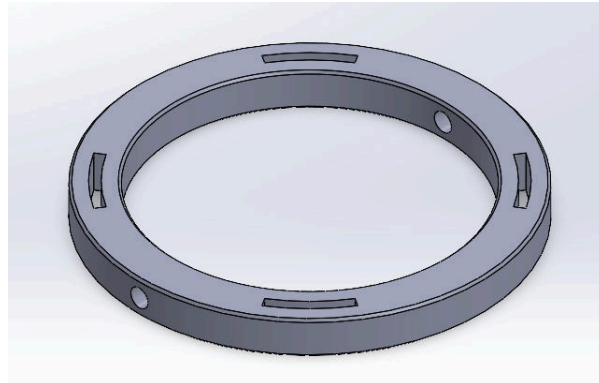
No part of the design proved delicate or prone to breaking. PLA provided strong, durable parts — durable enough that clearing printer residue from the pin-holes required noticeably more force than expected.

Process Improvements

The evaluation pointed to a clear fix: build tolerance into every mating fit rather than designing to a nominal dimension of zero clearance. Slightly increasing hole and fitment diameters would ease assembly without meaningfully affecting performance, and would target the smaller fitments — which caused more trouble than the overall assembly itself.

10. Design Improvements

Two changes were identified from the prototype evaluation: adding tolerance to mating fits, and adjusting ring sizing to improve the holder's range of motion. Enlarging the rings relative to one another — shrinking the inner ring while growing the outer ring — increases the achievable tilt angle.



Updated ring geometry with revised pin-hole and rib fitment tolerances.

Summary of Changes

The inner ring required the most rework. With original fits designed to zero tolerance, assembly was slow and required manual adjustment. The pin-hole diameter was increased from 0.175 in to 0.2 in, and the rib fitment was loosened by roughly 0.02 in in size and 5° total in fit angle (2.5° per side). A second prototype iteration incorporating these changes was not required and was not printed.

11. Reflection

Addressing Design Challenges

Once the zero-tolerance fits were identified as the source of tight assembly, the team sanded and lightly drilled the affected parts so they would seat comfortably, then re-tested each fitment. Where printer residue blocked a pin-hole, it was drilled out so the pin could seat properly.

Lessons Learned

Printing components individually — rather than all at once — proved far more effective, since it surfaced alignment issues early and allowed quick fixes without wasting print time or material.

Looking Ahead

Future iterations will build in tolerance and allowance from the start. Doing so should let parts fit comfortably rather than tightly, reducing both assembly rigidity and the post-print rework needed to make the design function as intended.

12. Project Links

Design files for this project are published to the following open platforms:

- Thingiverse (open-source repository): <https://www.thingiverse.com/thing:7208204>
- Thangs (e-commerce listing): <https://than.gs/m/1469641>